

Kondapalli Pranav Krishna

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LinkedIn | Portfolio

SUMMARY

Final-year Computer Science Engineering student specializing in Java, SQL, and MERN Stack with hands-on experience in building full-stack web applications and REST APIs. Strong foundation in OOP, backend systems, and JWT-based API security, with exposure to ML security and data poisoning attacks in autonomous systems.

SKILLS

Languages: Java, Python, C

Web: HTML, CSS, JavaScript, React

Backend: Node.js, Express.js, Spring Boot, Flask

Databases: MySQL, MongoDB

Core: OOP, REST APIs, JWT Auth, CRUD, SDLC

Tools: Git, GitHub, Thunder Client, Postman Agent, MATLAB

PROJECTS

Autonomous Vehicle Security – Data Poisoning Attacks (Capstone)

Oct 2025 – Apr 2026

Repo: GitHub Repository

- Designed an autonomous vehicle security pipeline to address vulnerabilities in perception systems using GTSRB, KITTI, and nuScenes datasets.
- Simulated attacks including label flipping, backdoor, clean-label poisoning, adversarial patches, LiDAR spoofing, GPS spoofing, and map poisoning.
- Implemented multi-layer defenses including Isolation Forest for anomaly detection, STRIP with Neural Cleanse and ULP for backdoor detection, PGD with median filtering, and GPS-IDS for sensor validation.
- Improved detection rate from 15% to 80% and reduced safety violations from 95% to 6.7% through integrated defense mechanisms.
- Validated performance using real-time CARLA simulation with fail-safe decision-making.

SHOPZILLA – Full-Stack E-Commerce (MERN)

Jun – Jul 2025

Live: Demo | Repo: GitHub

- Built and deployed a full-stack MERN e-commerce application using Netlify for frontend and Render for backend.
- Optimized backend APIs using caching and MongoDB query tuning, reducing average response time to ~245 ms (200 iterations, Postman).
- Evaluated authentication performance under load, achieving stable login response times of ~895 ms (100 iterations).
- Improved frontend performance via lazy loading and resource optimization, achieving Lighthouse score of 76 with FCP ~0.6s and LCP ~0.9s under Slow 4G.
- Monitored system and database performance using MongoDB Atlas and browser developer tools, ensuring stable API behavior.

PageForge BDP – Book Recommendation Engine & Demand Analytics System

Sep – Nov 2025

Live: Demo | Repo: GitHub

- Built a hybrid recommendation engine using TF-IDF vectorization and cosine similarity with preprocessing pipelines implemented using scikit-learn and pymongo.
- Implemented behavioral weighting using views, reviews, cart additions, and purchases to improve recommendation relevance and ranking quality.
- Designed interaction tracking and analytics pipelines using Precision@K, Recall@K, and CTR metrics to evaluate recommendation effectiveness and user engagement.
- Integrated Flask workflows with MongoDB-based interaction tracking for personalized recommendation generation and dynamic recommendation delivery.

EDUCATION

B.Tech in Computer Science Engineering, GITAM University

CGPA: 8.18

12th (TSBIE), PAGE Junior College

87.20%

10th (ICSE), International School

87.00%

CERTIFICATIONS

- Spring Boot for Beginners – Udemy
- IBM AI Engineering Professional Certificate – Coursera
- Python Essentials 1 & 2 – Cisco
- Cybersecurity Essentials – Cisco